

Machine Learning Analysis Report: Breast Cancer Diagnosis Classification

UCI Breast Cancer Wisconsin (Diagnostic) Dataset

Author: Timothy

Overview

This report addresses a supervised, binary classification problem: predicting whether a breast mass is malignant or benign from digitized measurements of cell nuclei. The dataset used is the Breast Cancer Wisconsin (Diagnostic) Data Set from the UCI Machine Learning Repository (<https://archive.ics.uci.edu/dataset/17/breast+cancer+wisconsin+diagnostic>). A logistic regression classifier was trained on 30 numeric features to predict the diagnosis label, and its performance was evaluated on a held-out test set.

Dataset Description

The dataset represents 569 fine needle aspirate (FNA) samples of breast masses, each digitized and analyzed to extract measurements of the cell nuclei present in the image (Street, Wolberg, & Mangasarian, 1993). It contains 569 rows and 32 columns: an identifier column (dropped before modeling), the diagnosis target (M = malignant, B = benign, occurring in 212 and 357 cases respectively), and 30 numeric features. Those 30 features are the mean, standard error, and largest ('worst') value of 10 underlying measurements -- radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, and fractal dimension -- describing the size, shape, and texture of the cell nuclei. The target variable, diagnosis, is what the model was trained to predict.

Modeling Approach

The `id` column was dropped since it is an arbitrary identifier with no relationship to diagnosis, and `diagnosis` was mapped to a numeric target (malignant = 1, benign = 0). The data was split into training (80%) and test (20%) sets using stratified sampling so both sets preserved the same malignant/benign ratio as the full dataset, and a StandardScaler was fit on the training set only and applied to both sets, to avoid leaking test-set information into preprocessing.

Logistic regression was selected because this is a binary classification problem with continuous numeric predictors, and logistic regression is a well-established, standard method for modeling exactly this kind of relationship while remaining directly interpretable through its coefficients (Hosmer, Lemeshow, & Sturdivant, 2013). Interpretability was a deliberate priority given the clinical context: a model whose coefficients can be inspected is easier to sanity-check against domain knowledge than an opaque one. ``class_weight="balanced"``` was used to compensate for the moderate class imbalance (63% benign, 37% malignant) in the training data.

Accuracy, precision, recall, F1-score, a confusion matrix, and an ROC curve with AUC were used to evaluate the model. Accuracy alone can be misleading on an imbalanced dataset, and different metrics answer different questions relevant to a diagnostic setting: recall on the malignant class measures how many true malignant cases the model catches, precision measures how many of its malignant predictions are correct, and the ROC/AUC summarizes class separation across all decision thresholds rather than only the default one; using multiple complementary metrics rather than accuracy alone is a widely recommended practice for classification evaluation (Powers, 2011). Logistic regression assumes a linear relationship between the (scaled) features and the log-odds of the outcome, and assumes the observations are independent of one another; both are reasonable working assumptions for this dataset, though the 30 features are themselves correlated with each other (they are mean/standard-error/worst versions of only 10 underlying measurements), which the model does not explicitly account for.

Results

On the 114-sample test set, the model achieved 97.4% accuracy, 97.6% precision and 95.2% recall on the malignant class, an F1-score of 0.964, and an ROC AUC of 0.995. The confusion matrix (Figure 1) shows 71 true negatives, 1 false positive, 2 false negatives, and 40 true positives -- only 3 of 114 test cases were misclassified, and only 2 malignant cases were missed. The ROC curve (Figure 2) sits very close to the top-left corner across nearly the entire threshold range, consistent with the near-perfect AUC. The coefficient plot (Figure 3) shows that ``texture_worst``, ``radius_se``, ``symmetry_worst``, and ``concave_points_mean`` had the largest positive influence on predicting malignancy, while features like ``compactness_se`` and ``fractal_dimension_se`` had the largest negative influence (pushing predictions toward benign).

Interpretation for a Non-Technical Audience

In plain terms, the model correctly identified the right diagnosis for 111 out of 114 test cases it had never seen before, including 40 of 42 malignant cases. That means, on this dataset, the model rarely confuses a malignant mass for a benign one, which is the type of error that matters most in a screening context. The measurements that mattered most to the model's predictions -- how irregular and large the cell nuclei were -- match what pathologists already look for when assessing whether a mass is cancerous, which is a reassuring sign that the model is picking up on medically sensible patterns rather than an unrelated quirk of the data.

Limitations and Potential Bias

One limitation is that this dataset comes from a single clinical source with 569 samples, which is small by modern machine learning standards and may not represent the full range of breast tissue presentations seen across different populations, imaging equipment, or labs; a model this size can look highly accurate on its own test set while still generalizing poorly to a different hospital's patients or equipment. A second limitation is that the 30 features are derived from only 10 underlying measurements (each reported as a mean, standard error, and worst value), so they are highly correlated with one another; this makes individual coefficients harder to interpret in isolation, since a change attributed to one feature could partly reflect a strongly correlated feature instead.

A potential source of bias is that this dataset's demographic and clinical context (for example, the patient population it was drawn from) is not documented in the version used here, so it is not possible to verify whether the model performs equally well across different patient subgroups. This matters because medical risk-prediction and diagnostic models have been shown to encode and even amplify demographic biases present in their training data or labels, sometimes in ways that are not obvious from aggregate accuracy figures alone (Obermeyer, Powers, Vogeli, & Mullainathan, 2019). Any real-world deployment of a model like this one would need subgroup-level evaluation and clinical validation before being used to support actual diagnostic decisions.

Reproducibility

Anyone can rerun this analysis by installing the pinned dependencies in requirements.txt with 'pip install -r requirements.txt' and executing modeling.ipynb top to bottom (for example, with 'python -m nbconvert --to notebook --execute modeling.ipynb'), using the included breast_cancer.csv file and a fixed random seed (random_state=42) for the train/test split and model training, which makes the reported results exactly reproducible.

References

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- Street, W. N., Wolberg, W. H., & Mangasarian, O. L. (1993). Nuclear feature extraction for breast tumor diagnosis. In *Biomedical Image Processing and Biomedical Visualization* (Vol. 1905, pp. 861-870). SPIE. <https://doi.org/10.1117/12.148698>